Eenheid 6.3

Note Title 2015/03/20

Die Puntproduk

Stewart p807

Definisie: Laat $\bar{a} = \langle a_1, a_2 \rangle$ en

$\bar{b} = \langle b_1, b_2 \rangle$ in $\mathbb{R}^2$

Ons definieer die puntproduk van $\bar{a}$ en $\bar{b}$ as

$$\bar{a} \cdot \bar{b} = \langle a_1, a_2 \rangle \cdot \langle b_1, b_2 \rangle$$
$$= a_1 b_1 + a_2 b_2$$

Antwoord is 'n Getal!!!

In $\mathbb{R}^3$ is $\bar{a} \cdot \bar{b} = \langle a_1, a_2, a_3 \rangle \cdot \langle b_1, b_2, b_3 \rangle$
$= a_1 b_1 + a_2 b_2 + a_3 b_3$

Antwoord is 'n getal!!

VOORBEELDE:

1) $\langle -2, 3 \rangle \cdot \langle 4, -5 \rangle = -8 - 15 = -23$

2) $(8\bar{i} - 2\bar{j} + 3\bar{k}) \cdot (2\bar{i} + 4\bar{k}) = \langle 8, -2, 3 \rangle \cdot \langle 2, 0, 4 \rangle$
$= 16 + 0 + 12$
$= 28$

3) $\langle 2, 3, -1 \rangle \cdot \bar{j}$
$= \langle 2, 3, -1 \rangle \cdot \langle 0, 1, 0 \rangle$
$= 0 + 3 + 0 = 3$

Eienskappe

Laat $\bar{a}, \bar{b}, \bar{c} \in \mathbb{R}^n$ en c is skalaar $(c \in \mathbb{R})$, dan

*1) $\bar{a} \cdot \bar{a} = |\bar{a}|^2$

2) $\bar{a} \cdot \bar{b} = \bar{b} \cdot \bar{a}$ ✓(kommutatiewe Wet)

3) $\bar{a} \cdot (\bar{b} + \bar{c}) = \bar{a} \cdot \bar{b} + \bar{a} \cdot \bar{c}$ ✓(Distributiewe Wet)

4) $(c\bar{a}) \cdot \bar{b} = c(\bar{a} \cdot \bar{b})$ ✓(assosiatiewe Wet)

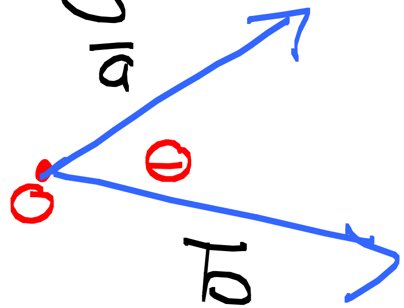
$\bar{a} \cdot (c\bar{b}) = c(\bar{a} \cdot \bar{b})$

*5) $\bar{0} \cdot \bar{a} = 0$ (antwoord is die getal 0)

$\Rightarrow \langle 0,0,0 \rangle \cdot \langle a_1, a_2, a_3 \rangle$

- <u>Definisie</u>

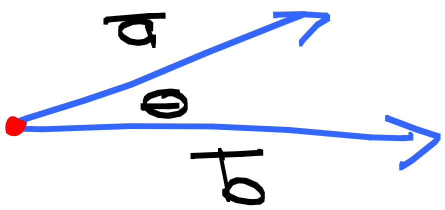
Laat $\bar{a}$ en $\bar{b}$ vektore wees in $\mathbb{R}^n$.
Die hoek θ tussen $\bar{a}$ en $\bar{b}$ word gedefinieer as die hoek tussen die voorstelling wat by die oorsprong begin en $0 \leq \theta \leq \pi$



Let op: As $\bar{a}$ en $\bar{b}$ parallel is dan is $\theta = 0$ of $\theta = \pi$

Stelling

Laat θ die hoek tussen $\bar{a}$ en $\bar{b}$ wees,



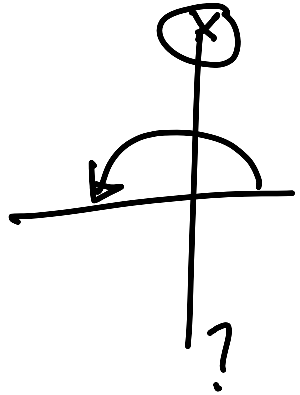
dan is $\bar{a}\cdot\bar{b} = |\bar{a}||\bar{b}|\cos\theta$

As $\bar{a} \neq \bar{0}$ en $\bar{b} \neq \bar{0}$ dan

- $\cos\theta = \dfrac{\bar{a}\cdot\bar{b}}{|\bar{a}||\bar{b}|}$ $\qquad \theta \in [0,\pi]$

Voorbeelde ① Bepaal die hoek θ tussen $\bar{a} = \langle -1,1\rangle$ en $\bar{b} = \langle 1,1\rangle$

$$\cos\theta = \frac{\langle -1,1\rangle\cdot\langle 1,1\rangle}{(\sqrt{2})(\sqrt{2})} = \frac{-1+1}{2} = \frac{0}{2} = 0$$



$\therefore \cos\theta = 0 \Rightarrow \theta = \pi/2$

② Bepaal die hoek θ tussen $\bar{a} = \langle 1,-2,3\rangle$ en $\bar{b} = \langle -3,2,1\rangle$

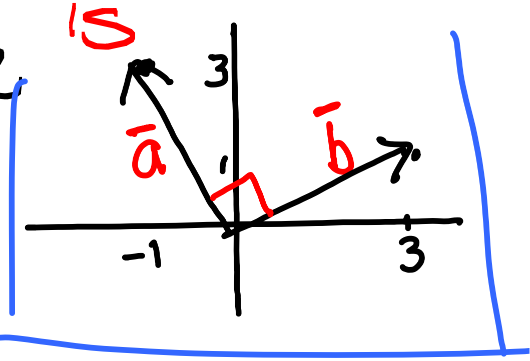
$$\cos\theta = \frac{\langle 1,-2,3\rangle\cdot\langle -3,2,1\rangle}{(\sqrt{14})(\sqrt{14})} = \frac{-3-4+3}{14} = -\frac{2}{7}$$

$$|\bar{b}| = \sqrt{(-3)^2+(2)^2+1^2}$$

$$\therefore \theta = \text{bg}\cos\left(-\frac{2}{7}\right)$$

Definisie Twee nie-nul vektore $\bar{a}$ en $\bar{b}$ is ortogonaal (loodreg) as die hoek tussen die twee vektore $\theta = \pi/2$ is

Bv. $\bar{a} = \langle -1,3\rangle$ en $\bar{b} = \langle 3,1\rangle$



$\bar{a}\cdot\bar{b}$

$$\cos\theta = \frac{\langle -1,3\rangle\cdot\langle 3,1\rangle}{|\bar{a}|\,|\bar{b}|} = \frac{-3+3}{10} = 0$$

$$\therefore \theta = \pi/2$$

Dus: Twee nie-nul vektore is ortogonaal $(\bar{a} \perp \bar{b})$ as en slegs as $\bar{a} \cdot \bar{b} = 0$

VOORBEELD: Bepaal k sodat $\bar{a} = \langle -1, 3, k \rangle$ en $\bar{b} = \langle 2, k, 3 \rangle$ ortogonaal is.

$$\bar{a} \perp \bar{b} \Rightarrow \bar{a} \cdot \bar{b} = 0$$

$$\Rightarrow \langle -1, 3, k \rangle \cdot \langle 2, k, 3 \rangle = 0$$

$$\Rightarrow -2 + 3k + 3k = 0$$

$$\Rightarrow 6k = 2$$

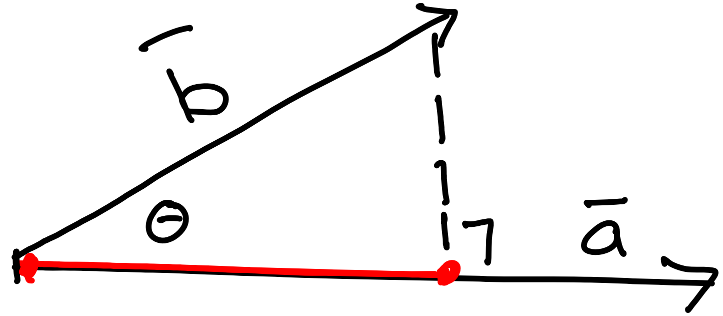
$$\Rightarrow k = \frac{2}{6} = \frac{1}{3}$$

Skalaar en Vektorprojeksies P 811 Stewart

No 7 van Eenheid 6.3 in Studiegids

Twee projeksies van 'n vektor $\bar{b}$ op 'n vektor $\bar{a}$.

Skalaar projeksie



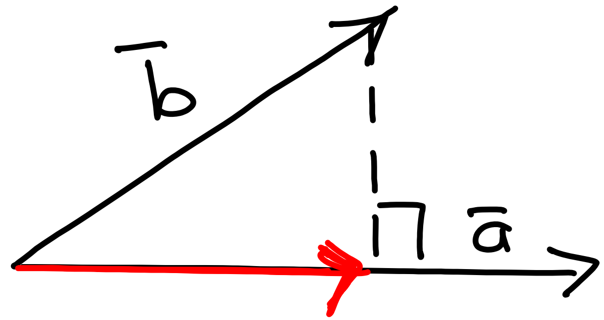
$$\text{komp}_{\bar{a}}\bar{b} = |\bar{b}|\cos\theta$$

$$= \cancel{|\bar{b}|} \times \frac{\bar{a}\cdot\bar{b}}{|\bar{a}|\cancel{|\bar{b}|}}$$

$$= \frac{\bar{a}\cdot\bar{b}}{|\bar{a}|}$$

(antwoord is 'n getal)

Vektor projeksie



$$\text{proj}_{\bar{a}}\bar{b} = \text{komp}_{\bar{a}}\bar{b} \times \frac{\bar{a}}{|\bar{a}|}$$

$$= \frac{\bar{a}\cdot\bar{b}}{|\bar{a}|} \times \frac{\bar{a}}{|\bar{a}|} = \frac{\bar{a}\cdot\bar{b}}{|\bar{a}|^2} \times \bar{a}$$

(antwoord is 'n vektor)

Voorbeeld:

Gegee $\bar{a} = \langle 1,2,3 \rangle$ en $\bar{b} = \langle -1,0,2 \rangle$ vind die skalaar en vektor projeksies van $\bar{b}$ op $\bar{a}$.

$$|\bar{a}| = \sqrt{1^2+2^2+3^2} = \sqrt{14}$$

1) Skalaar Projeksies

$$\text{Komp}_{\bar{a}}\,\bar{b} = \boxed{\frac{\bar{a}\cdot\bar{b}}{|\bar{a}|}} = \frac{\langle 1,2,3\rangle \cdot \langle -1,0,2\rangle}{\sqrt{14}}$$

$$= \frac{-1+0+6}{\sqrt{14}} = \frac{5}{\sqrt{14}}$$

(getal)

(2) Vektor Projeksies

$\bar{a} = \langle 1, 2, 3 \rangle$

$\bar{b} = \langle -1, 0, 2 \rangle$

$$\text{proj}_{\bar{a}}\, \bar{b} = \frac{\bar{a} \cdot \bar{b}}{|\bar{a}|^2} \times \bar{a} \quad ??$$

$$= \frac{5}{(\sqrt{14})^2} \times \langle 1, 2, 3 \rangle$$

$$= \left\langle \frac{5}{14}, \frac{10}{14}, \frac{15}{14} \right\rangle$$

Los no 6 van Eenheid 6.3

Rigtingshoeke en Rigtings cosinus.